# Observational Reboot: Conditional SMICA Pilot and Research Programme

HTT observational research

14 October 2026

**Abstract**

We executed a source-bound, retrospective exploratory low-mode SMICA experiment, a finite-null calibration utility, and a local/global response laboratory. The global rank is $129/241 = 0.535270$. It is conditional on a declared delivered-simulation model, not a confirmatory physical Planck null probability or a Bianchi result. The continuation covers all feasible analyses in the existing theory, code and data.

## 1 Executed SMICA experiment

The K1 NSIDE64 cache was verified against its accepted byte-identity gate. One observation and 300 separately delivered signal/noise pairs use the identical 38,413-pixel mask and simultaneous 36-column $\ell = 0, \ldots, 5$ fit. Each simulation component is used once. Training uses 60 pairs; calibration uses the other 240. The four features are squared norms of retained real harmonic coefficients divided by $2\ell + 1$. These are selected-sky fitted power summaries in $\mu\mathrm{K}^2$, not certified unbiased full-sky power estimates.

| $\ell$ | Power summary | Marginal rank | Max-adjusted rank |
|---|---|---|---|
| 2 | 226.451 | 0.1120 | 0.5353 |
| 3 | 402.198 | 0.5353 | 0.9461 |
| 4 | 239.881 | 0.5685 | 0.9751 |
| 5 | 291.639 | 0.2697 | 0.6971 |

The score is the maximum of four absolute departures standardized using training only. There are 128 exceedances among 240 calibration rows, giving $129/241 = 0.535270$. The rank floor is $1/241$. The 95% Clopper–Pearson interval $[0.468058, 0.597775]$ concerns the IID conditional null-tail probability, not the realized discrete rank.

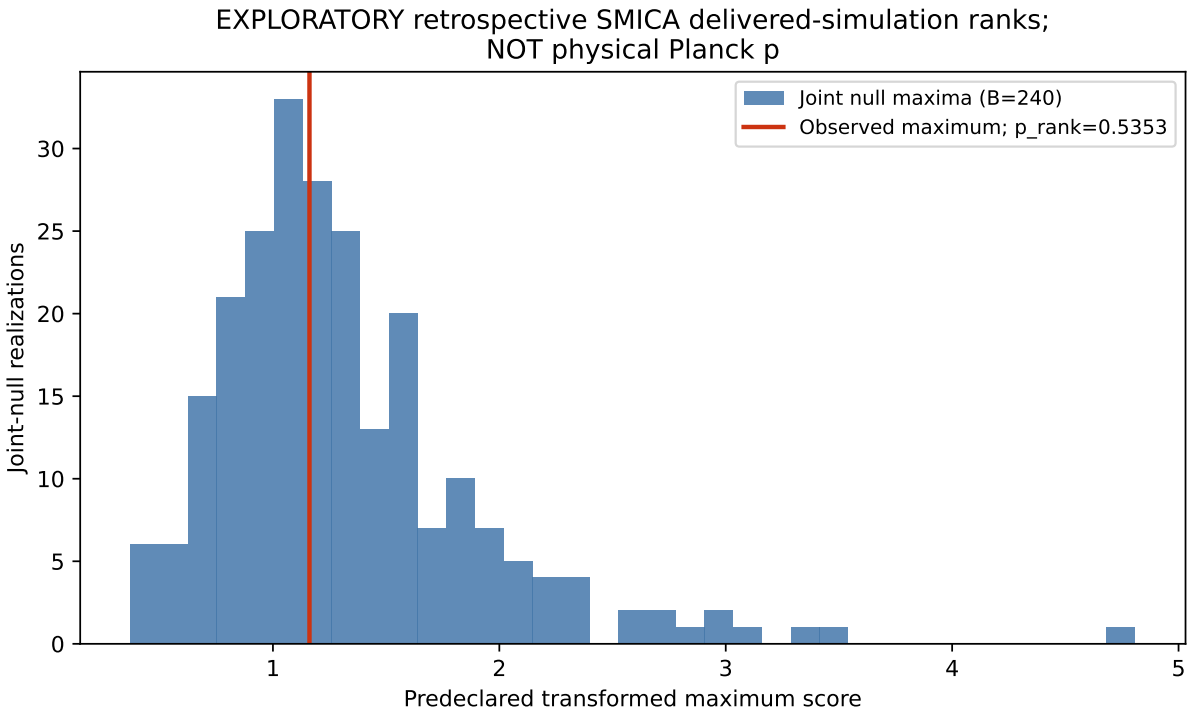



Figure 1: Actual SMICA, retrospective conditional delivered-simulation rank. The feature family and current split were fixed before this run, after historical examination of this sky.

## 2 Interpretation and historical comparison

Planck's release describes independent noise/systematic simulations and linear separate propagation at fixed actual-data weights. The official plancklens library adds delivered signal and noise maps. This supports our addition recipe, while observed-simulation exchangeability, foreground residuals, fixed weights and systematic means remain physical-model obligations. We did not subtract the simulation mean as an assumed observed bias. The OLS identity residual near $2.1 \times 10^{-15}$ verifies algebra, not omitted-mode leakage.

The old K1 rank 0.039 used another processing/statistic family and 999 signals with 300 cyclically reused noise draws. Archive 0.078/0.079 were posterior-predictive discrepancies combining dipole, CF4 and scalar D2/D3 summaries. The R8 finite-rank enclosures were synthetic pools. None is an interchangeable predecessor of this result. The legacy review also identifies $D_\ell/C_\ell$ and STF normalization differences.

## 3 Finite-null implementation checks

A fixed-$B = 999$, 1000-repetition correlated-feature experiment returned a 0.060 global rejection rate; its 95% binomial interval is [0.04610,0.07656]. Choosing the smallest unadjusted marginal rank returned 0.145. The campaign is conditional on one fixed independent training sample. This tests the declared synthetic procedure, not empirical Planck calibration. Tests exercise exact inclusive ties, binomial interval coverage, disjoint training, same-sky rejection and projection.

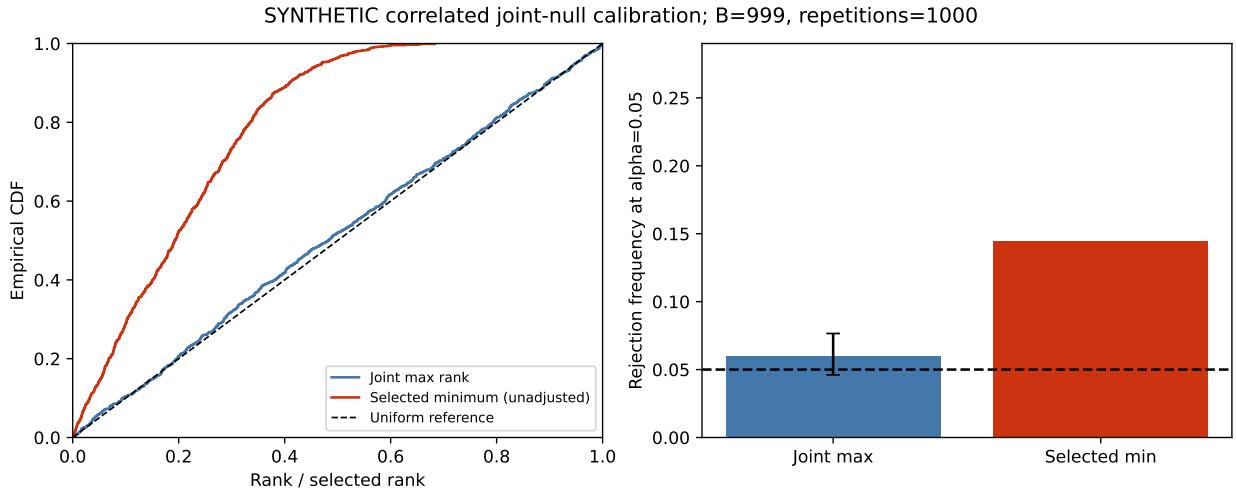



Figure 2: Synthetic calibration experiment. Error rates apply to the specified correlated Gaussian generator; the Planck physical law is a separate obligation.

# 4 Local/global response and continued analysis

For a declared linear Gaussian model $y = B\beta_L + G\beta_G + \epsilon$, nuisance projection after covariance whitening gives the conditional information $G_w^T(I - P_{B_w})G_w$. In the synthetic laboratory the constant response has rank 0, the specified depth response has rank 3, and free shell nuisance gives rank 0. Full rank can still be arbitrarily weakly informative. Nuisance-column normalization preserves its span under parameter-unit changes. The theory note derives endpoint boost/STF response, a nuisance quotient, two-shell

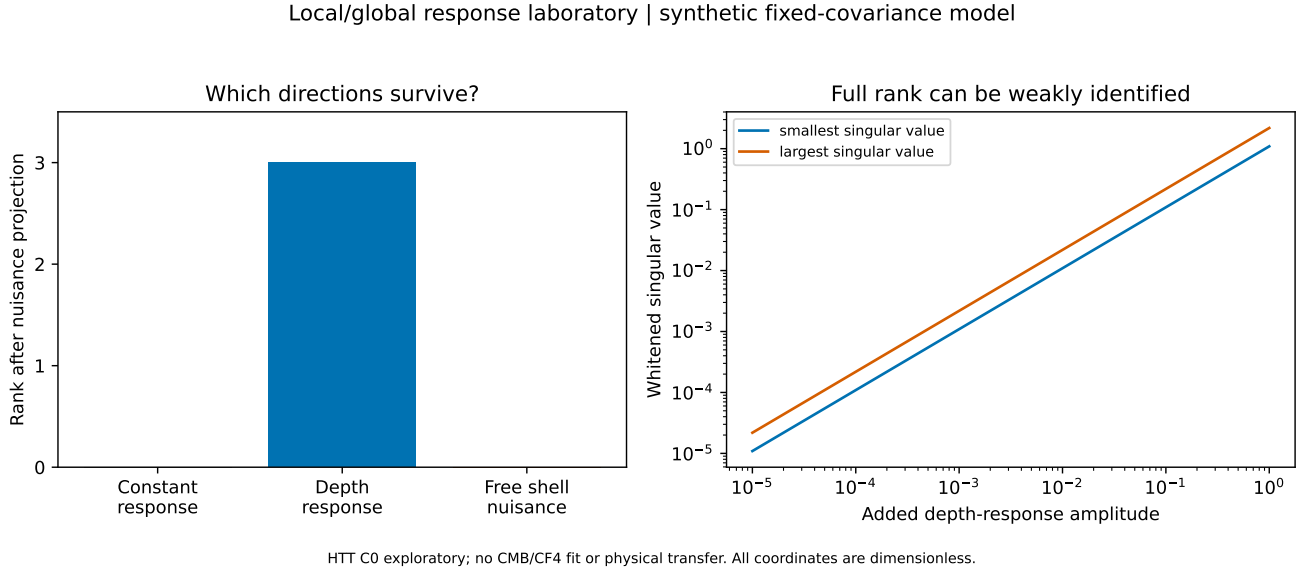



Figure 3: Specified dimensionless response model with known covariance. This is not an observed velocity or global-tilt estimate.

discrimination, shared-sky covariance and a restricted Bianchi-I history. Small Wolfram and NumPy checks are not four-axis CAS admission or a native atlas.

The 23-card DAG and 20-opportunity inventory cover SMICA, physical dipole and local/global response, CF4/SDSS depth laws, DESI/Union3/JWST distances, released lensing and spectra, and polarization controls. Codex must assign concrete goals, execute and review each ready analysis, publish generated figures/reports and continue. Missing native inputs block their branch only. Capability discovery continues beyond this sampled inventory; the present completed unit is RB00.

## Source and reproduction pointers

Full numerical input hashes and source paths: `report.json`. Commands, failures and targeted repairs: the sibling `evidence` directory. Independent review, goal queue and copyable handoff are linked from the package README. Primary source context: Planck 2018IV, arXiv:1807.06208; Planck 2018VII, arXiv:1906.02552; Dai–Chluba, arXiv:1403.6117; Roldan et al., arXiv:1603.02664; Saadeh et al., arXiv:1605.07178. See the source registry for exact read scope and methodological limits.